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For OMI_CIRCULATION_VOLTRANS_ARCTIC_averaged

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

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1) INTRODUCTION

a) Product Name

OMI_CIRCULATION_VOLTRANS_ARCTIC_averaged

b) Indicator Evaluation Method

The Atlantic Water volume transport time series provided here represents the integrated ocean volume transport through the Faroe-Shetland Channel, Barents Sea Opening and Fram Strait. The indicator timeseries are validated using in-situ based observations found in literature. For details, see the Ch. 3.2 of the Ocean State Report #3 (von Schuckmann et al., 2018). Here, Atlantic Water is defined using the following temperature criteria: T > 5 °C (Faroe-Shetland Channel); T > 3 °C (Barents Sea Opening); T > 2 °C (Fram Strait). The locations of the sections are shown in Figure 1. The temperature criterion applied excludes any transport of colder water masses of Arctic origin in the northern parts of the sections. Moreover, most of the water masses transported by the Norwegian Coastal Current in the southern part of the section are also excluded. The volume transport is expressed in Sverdrups (Sv; 1 Sv = 10^6 m^3s^{-1}).
The overflow waters volume transport time series provided here represents the integrated ocean volume transport through the Faroe-Shetland Channel and the Denmark Strait that meet the density criteria (σ_θ > 27.8 kg m^{-3}).

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
ARCTIC_MULTIYEAR_PHY_002_003	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-ARC-QUID-002-003.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-ARC-PUM-002-003.pdf
ARCTIC_ANALYSISFORECAST_PHY_002_001	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-ARC-QUID-002-001.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-ARC-PUM-002-001.pdf

2) SCIENTIFIC VALIDATION

a) Nordic Seas volume transport

The positions of the sections in the reanalysis data are taken such that it matches the location of standard hydrographic sections that are also monitored by arrays of current meter measurements. In the Nordic Seas, two sections which are comparatively well-monitored are the Faroe-Shetland Channel (FSC) at the southern entrance to the Nordic Seas and the Barents Sea Opening (BSO) between the Norwegian and Barents seas. The flow of Atlantic Water through the FSC has been estimated to 2.7 Sv, with a standard deviation of 1.1 Sv, for the period 1995 to 2015 (Østerhus et al., 2019). The corresponding model-based estimate is 2.3 Sv for the full period 1993-2024. Thus, the model tends to be on the lower side compared to the observations. The observations indicate a trend of -0.006 Sv/year for the period 1993-2015, but the trend is not statistically significant (Østerhus et al., 2019). The model results, on the other hand, indicate a trend of -0.01 Sv/year for the full period 1993-2024.
It should be noted that the observation estimates are based on a combination of an ADCP (Acoustic Doppler Current Profiler) array with configurations changing somewhat in time, and current estimates from satellite altimetry remote sensing observations (Berx et al., 2013). Moreover, the observations represent the Atlantic Water part of the flow through the Faroe-Shetland Channel, which is characterized by both northward inflow and southward recirculation of Atlantic Water (e.g., Berx et al., 2013). The model estimates are based on the net northward flow of water masses where the temperature $T > 5$ °C across the full section. Hence, a direct comparison between observation-based and model-based estimates should be interpreted with caution.

Downstream in the BSO, the model indicates a long-term (1993-2024) average net Atlantic Water inflow of 2.2 Sv, with a long-term positive trend of 0.003 Sv/year. This estimate is a bit on the high side compared with observations, with an estimated average net inflow of around 2 Sv, as reported previously in Smedsrud et al. (2013). This estimate was based on observation data from the period 1997-2010. Moreover, an additional 1.2 Sv is carried by the Norwegian Coastal Current, which is situated along the Norwegian coast. In the model data, the Norwegian Coastal Current is excluded through a geographical boundary at 71°30'N, which approximately represents the boundary between the two current inflows and also represents the position of the southernmost current meter mooring used to monitor the Atlantic Water inflow (see e.g., Ingvaldsen et al., 2004). Note, however, that also in this section the current meter mooring array has been modified somewhat through the years (see e.g., Ingvaldsen et al., 2007). The model data indicate a positive trend of 0.003 Sv/year over the period 1993-2024. While no trend has been reported in the literature based on observation data, a recent model study has indeed found a positive trend in the Atlantic Water inflow through the BSO (Jahanmard et al., 2025).

The part of the Atlantic Water flow towards the Arctic Ocean that does not enter the Barents Sea, continues northwards along the west coast of the Svalbard archipelago as the West Spitsbergen Current. In the model, the Atlantic Water volume transport through the Fram Strait is calculated as the net transport of water masses where $T > 2$ °C. This yields an average volume transport of 1.2 Sv, with a positive trend of 0.03

Sv/year. For comparison, an average, nearly constant volume transport of 1.3 Sv has been estimated for the Svalbard branch of the West Spitsbergen Current, based on an array of moored current meters in the period 1997-2010 (Beszczynska-Möller et al., 2011). The currents through the Fram Strait show a complex structure with several bands of northward and southward flow, which makes it challenging to estimate accurate volume transports. Moreover, the choice of a temperature criterion further complicates the direct comparison between model and observations with respect to both mean and variability, because model biases in temperature will affect the model transport estimates.

One main constituent of the return flow from the Nordic Seas to the North Atlantic Ocean is cooled water with density $\sigma_\Theta > 28.0$ (i.e., a density exceeding 1028.0 kg/m^3). Two main branches exit the Nordic Seas through the Faroe-Shetland Channel and through the Denmark Strait.
The modelled densewater overflow through the Faroe-Shetland Channel amounts to 0.6 Sv as an average over the period 1993-2024, with a trend of 0.01 Sv/year. This is considerably smaller than the 1.9 Sv reported by Hansen and Østerhus (2007), based on ADCP measurements during the period 1995-2005. The measurements were obtained further downstream in the Faroe Bank Channel, and may therefore be somewhat enganced due to entrainment while en-route. Still, the model-based estimates appear to be too small in comparison to the observation-based estimates. According to Hansen et al. (2016), the overflow through the Faroe-Shetland Channel appears to be stable over time.
The largest overflow of densewater from the Nordic Seas occurrs through the Denmark Strait. Here, the model shows an average overflow of 1.3 Sv, with a negative trend of -0.003 Sv/year, for the period 1993-2024. This is again small compared to observation-based estimates suggesting an average overflow of 3.4 Sv during the period 1996-2011 and with no significant trend (Jochumsen et al., 2012).

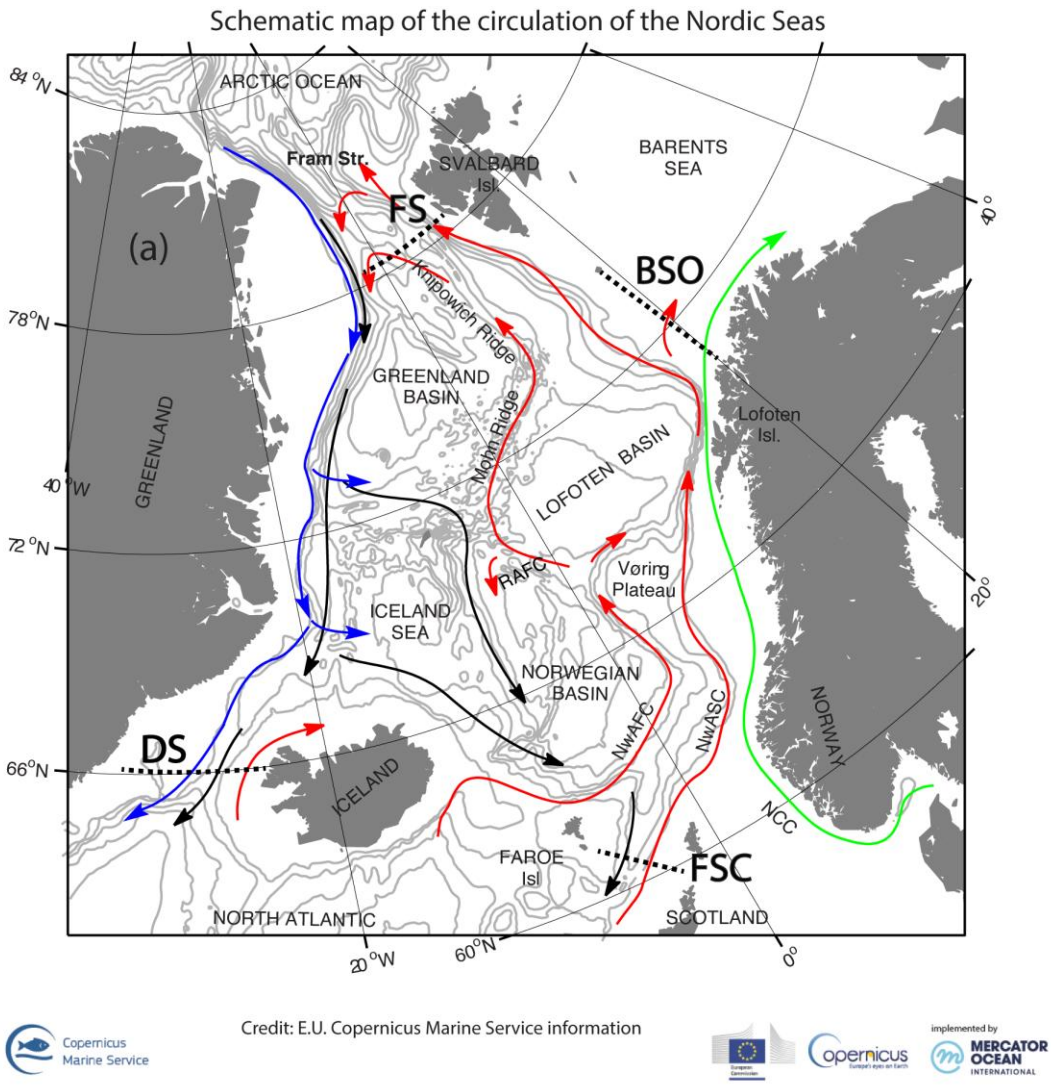

Figure 1: *The Nordic Seas with schematic water pathways showing its overturning circulation from northward flowing Atlantic Water in the surface (red) to southward flowing transformed waters at depth (black). The two branches of the Norwegian Atlantic Current, the Norwegian Atlantic slope current (NwASC) and Norwegian Atlantic front current (NwAFC) are represented by red arrows. The fresh Norwegian Coastal Current (NCC) is indicated in green, while the Greenland Current in marked in blue. See Raj et al., 2016 for details. Grey isobaths are drawn for every 600 m. The dashed lines provide the locations of the sections; Faroe Shetland Channel (FSC), Barents Sea Opening (BSO), Fram Strait (FS) and Denmark Strait (DS).*

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